

Does Product Categorization Granularity Affect the Functionality of Diffusion Metrics in the Product Space? A Re-Examination of Hidalgo et al. (2007)

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① Discuss what the shift from 4-digit to 3-digit means for aggregation.



you argue for more variance but you decrease granularity

② special granularity? ← ~~new~~ economic geography

③ impact: data problem in many contexts

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Research Problem

The Product Space Conditions the Development of Nations article by Hidalgo et al. (2007) introduces a network representation of the relatedness between globally traded products, called the product space. Their model is based on the Standard International Trade Classification revision 4 (SITC-4) system, and each node represents a product class with edges connecting product classes frequently co-exported by the same country. The authors intend that this network would act as a roadmap for nations seeking to shift their focus of production toward higher-value goods by highlighting likely paths for diffusion.

However, since categorization is inherently subjective, different categorization systems and levels of granularity exist. For example, songs could be categorized by genre, tempo, key, rhythm, energy, and many others. The characteristic which defines the classification of songs significantly influences the resulting partition. The same applies to any categorization, including the SITC-4 system utilized by Hidalgo et al. (2007). Thus, the question, “What is the most useful categorization for the product space described by Hidalgo and his co-authors (2007)?” follows from this logic. Two important factors should be considered here, the first being functionality; products within the same class should require similar inputs, otherwise proximities would misrepresent the potential for diffusion through the product space. Second, categorization granularity must be consistent throughout the product space as ease of diffusion from one section to another would be distorted by resolution inconsistencies.

*
SITC
products
↓
↑
spatial
units

We hypothesize that the diffusion patterns described by Hidalgo et al. (2007) might have looked different if the researchers used a different classification system or level of granularity. Was the level of resolution sufficiently granular, or would a coarser system result in similar conclusions? An analysis of the differences between categorization systems, and specifically the granularity levels within these systems, would allow a full view of the relationships within the product space to enhance the complexity and rigor of the Hidalgo et al. (2007) study. This analysis aims to study the effect of system granularity on proximity and ease of diffusion.

Description of Data Collection

Because we wish to analyze and compare granularities of the same data used in the original paper, the dataset we used is from the NBER-United Nations (bilateral) Trade Data 1962-2000, accessed through the authors' supporting online materials document. The

* Hidalgo 2007 SI robustness
Zaccaria et al. 2014 PLOS ONE
Pikerno

O'Leary 2012 provides
algorithm for resolution-
selection

variables of interest from this dataset, aligned with the same variables of interest from the Hidalgo et al. (2007) paper, were the countries who exported the products (“exporter”), the SITC revision 4 (SITC-4) codes for the product classes exported (“sitc4”), and the value of the total exports for the year for each country in given sitc4 category (“value”). We used the datasets from 1995-2000 including the time period from 1998-2000 from the Hidalgo et al. (2007) paper as well as a comparison time period from 1995-1997.

Methodology

Our aim is to analyze how changing categorization granularity distorts both the structure of the product space and the ease of diffusion through it. We define distortion as significant changes in average proximity and diffusion potential when moving between time periods and levels of granularity. We will compare the level 3 and 4 categories from the SITC-4 system for the sets of years 1995-1997 and 1998-2000. The latter matches the original Hidalgo et al. (2007) study, and offers a comparable baseline. The SITC-4’s level four categorization includes specific products (eg. 8417: “Electric Motors and Generators”), while level three defines broader industry groups (eg. 841: “Machinery for Industrial Use”).

Data Preprocessing

Using the original paper’s supporting online materials as a guide, we recreated the dataset used by downloading trade data from the years 1998 and 2000 and aggregating the export values across all three years. We further aggregated the exports by destination country, resulting in one total export value per product class per country. We then removed all product codes not present in the Standard International Trade Classification (SITC) system. The final dataset contained 775 SITC-4 categories – the same number reported in the Hidalgo et al. (2007) paper. This suggests a successful replication, as the full SITC-4 classification includes a total of 1,023 possible four-digit categories.

To generate the three-digit version, we removed the last digit of each SITC value from our finalized level four data frame and aggregated all observations with the same SITC value. This process was repeated for the years 1995-1997.

Classification System Comparison

To compare granularity levels, we will first compute the proximity matrices for each of the four data frames (level 3 for 1995-1997, level 3 for 1998-2000, level 4 for 1995-1997,

level 4 for 1998-2000). We will calculate the Revealed Comparative Advantage (RCA) values for each granularity level and time period for each country c and good i :

$$RCA_{c,i} = \frac{x(c,i)}{\sum_i x(c,i)} \bigg/ \frac{\sum_c x(c,i)}{\sum_{c,i} x(c,i)}$$

and utilize these values in the authors' formula for Proximity (ϕ)

$$\phi_{i,j} = \min\{P(RCAx_i|RCAx_j), P(RCAx_j|RCAx_i)\}$$

for each pair of products i and j within each categorization and time period. Using R, we aggregate these values into four proximity matrices, one for each of the granularity levels and time periods.

We will then create matrix visualizations for each proximity matrix in the same fashion as the authors' (Hidalgo et al., 2007) figure 1a to understand the data and relationships between products in the product space. In the Hidalgo et al. (2007) paper, the proximity matrix representation in figure 1a showed that 5% of the elements were equal to zero, 32% were smaller than 0.1, and 65% took values lower than 0.2 (for level 4 in 1998-2000). We will compute these values for our other three data frames of interest to compare overall proximity of products in the product space. From these visualizations and their matrices, we will analyze changes in mean proximity and the share of $\phi > 0.55$ links for the level 3 versus level 4 granularity, and the 1995-1997 versus 1998-2000 time periods.

To illustrate that, "...countries develop RCA in goods close to those they already had is to calculate the conditional probability of transitioning into a product given that the nearest product with $RCA > 1$ is at a given proximity level (ϕ) (Hidalgo et al., 2007), we will use test countries Chile and Korea to simulate diffusion in the product space as these were displayed as simulation examples in Hidalgo et al. (2007). If we have the time and necessary resources, we would like to create the same network visualizations as in figure 4A from the paper which

show the number of steps from one product to another, and in our version, we will create these networks for both sets of years and both levels of categorization. Following the process outlined in the *Simulated Diffusion* section of Hidalgo et al.'s *Supporting Online Materials* document, we will first identify the products which Chile and Korea export for the years 1998-2000 in the fourth level of granularity, to verify our simulation process with Hidalgo et al. (2007). Then, we will identify all products which are within a certain proximity (ϕ) from their current exports and include these in their list of exports – this is considered to be one iteration of the simulation. We will count the number of reachable products after each iteration (20 total), and compare these results between granularity levels 3 and 4. Finally, we will repeat this process and compare results for time period 1995-1997.

Expected Outcomes

For all time periods, we expect that the level 3 data frames' proximity matrices show less proximity overall due to the disappearance of more granular, shorter ties. We expect these matrices to have a “core-periphery” structure as well, though less intense in the level 3 networks. We expect that percentage of proximities for values less than 0, less than 0.1 and less than 0.2 will be greater than 5%, 32% and 65%, respectively at granularity level three, for the same reason as above. We expect that shares of proximity values will be greater in lower value bins in the years 1995-1997 than 1998-2000 as countries tend to produce more exports with time, so their proximities will likely increase as time passes.

In Hidalgo et al.'s paper, both Chile and Korea were able to diffuse to the center of the product space when $\phi > 0.55$ (with Korea able to diffuse much quicker), but with a less granular categorization at the 3-digit level, we expect that both countries will take more time to diffuse to the center, with Chile possibly not being able to diffuse after 20 iterations. We also expect that diffusion in the 1995-1997 time period will be slower and more difficult than that of the 1998-2000 time period as, with increased amounts of exports over time typically comes with decreased proximities to other products, and thus easier diffusion.

Impact

This paper analyzes whether the methods used in Hidalgo et al. (2007) generalize to coarser levels of product categorization and how granularity affects proximity measures and diffusion outcomes. In doing so, we demonstrate how variances in classification granularity can yield different results and interpretations, highlighting the importance of careful

categorization system selection. This paper can be useful for policymakers who risk overgeneralizing development pathways with coarse levels of classification, and can help to guide more informed investment strategies which delegate resources more efficiently in the pursuit of economic growth. This paper also offers a replicable template for future researchers when comparing categorization systems like the Harmonic System (HS) or the North American Industry Classification System (NAICS), or testing different granularity levels over time. Finally, for the original paper, this insight could inspire reform to the authors' original methods, expanding their scope and reconsidering the basis of their research, encouraging more nuanced results.

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